into some quite intense discussions.” Now, surrounded by the unfamiliar and stark California landscape, Bell set about doing just that. In the process, he ended up discovering a remarkable truth about the quantum world—and ultimately loosened the grip of the Copenhagen interpretation on the collective psyche of physics.

John Stewart Bell was born on June 28, 1928, the second of four children in a working-class Protestant family in Belfast, Northern Ireland. By his own account, he came from a long line of “carpenters, blacksmiths, laborers, farm workers, and horse dealers.” Bell was the first in his family to attend high school—his father had left school at the age of eight, and his siblings all found jobs by fourteen. By sixteen, Bell had graduated from the least expensive high school in the area, but the local university, Queens, wouldn’t admit anyone younger than seventeen. So he went looking for work. “I applied to be office boy in a small factory, some starting job at the BBC—things like that. But I didn’t get any of the jobs I applied for,” Bell recalled years later. Ultimately, he found work as a lab assistant in the physics department at the university. “It was a tremendous thing for me, because there I met, already, my future professors. They were very kind to me. They gave me books to read, and in fact, I did the first year of my college physics when I was cleaning out the lab and setting out the wires for the students.”

Toward the end of his formal studies at Queens, Bell had his first encounter with the mathematics of quantum physics, and the Copenhagen interpretation that invariably accompanied it. He was not happy with what he found. “You learn about the periodic table of the elements—all the practical aspects of the theory,” Bell recalled. “Then the puzzles start.” Bell’s instructors and textbooks were vague about the nature of the wave function itself. They were never clear “whether it [the wave function] was something real or some kind of bookkeeping operation.” And if the wave function was just a bookkeeping device, just information, then whose information was it? And if there really was no quantum world, as Bohr had insisted, then what was that information about? Bell even got into an